

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	[Enter Team ID]
Project Name	HDI Predictor
Maximum Marks	5 Marks

Functional Features Overview

This document lists all the functional features implemented in the HDI Predictor project. Each feature contributes to automating the country human development index calculation and prediction process using Machine Learning and a Flask web application.

Functional Features

S.No	Feature Name	Feature Description	Module / Component	Status (Done / in progress / Pending)	Marks Contribution
1	Indicator Data Input	Accepts country indicator details through a web form.	Flask UI	Done	High
2	Data Preprocessing	Cleans, scales, and prepares input data for prediction.	Data processing	Done	High
3	HDI Score Prediction	Predicts a country's HDI score and development tier.	Prediction Engine	Done	Medium
4	Model Integration	Loads the trained model into the Flask application using Pickle.	Backend	Done	Medium
5	User-Friendly Web Interface	Displays prediction results instantly through an interactive web page.	Frontend	Done	Medium

Feature Summary:

Metric	Count / value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Indicator Data Input, Data Preprocessing, Model Training, HDI Prediction
Additional / Nice-to-have Features	User-Friendly Interface, Error Handling, Model Serialization
Features Tested & Verified	8 Features Successfully Tested

Feature Category Breakdown

S.No	Category	Features in Category	Example Feature
1	User Interface(UI)	2	Indicator Information Form, Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, Prediction Engine, Error Handling
3	Database / Storage	1	Dataset Management and Pickle Model Storage
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	1	Input Validation and Exception Handling